# GAT-assisted Wireless Link Scheduling for Dense UAV Swarm Networks

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***Abstract*—In dense uncrewed aerial vehicle (UAV) swarm networks, massive node connectivity leads to severe interference and collisions, intensely limiting network resource utilization. Therefore, it is imperative to achieve efficient and distributed wireless link scheduling. Recently, graph neural networks (GNNs) have emerged as a promising paradigm for wireless scheduling problems. However, existing GNN-based methods often rely on simplified network modeling or global graph computation, making them less suitable for densely connected, fully distributed and energy constrained UAV ad hoc networks. To address these challenges and promote green communications, this paper proposes a distributed scheduling framework via graph attention networks (GATs). We first construct a weighted directed conflict graph to capture heterogeneous and asymmetric interference relationships. An edge-enhanced attention mechanism is then introduced, where edge weights are explicitly incorporated into the attention mechanism, enabling each link to infer dominant interference pattern from limited neighborhood interactions. Simulation results demonstrate that the proposed scheme significantly outperforms conventional benchmarks in terms of network throughput, while exhibiting high data efficiency, fast convergence and strong scalability in dense UAV swarm scenarios. It dramatically reduces the computational energy consumption, making it suitable for green and sustainable dense UAV swarm scenarios.**

***Index Terms*—Wireless link scheduling, Graph attention networks, UAV swarm networks, Interference management.**

## I. Introduction

Driven by the rapid evolution of 6G communication and low-altitude economy, uncrewed aerial vehicles (UAVs) have attracted increasing interest to support diversified tasks. To handle complex collaborative missions, flying ad-hoc networks (FANETs) have emerged as a preferable architecture due to their decentralized, self-organizing, and resilient characteristics [1]. However, as swarm density grows, limited wireless resources lead to mutual interference and packet collisions, sharply degrading overall network throughput and spectrum efficiency, while causing severe energy wastage due to retransmissions.

In dense FANETs, traditional contention-based access such as CSMA/CA [2] becomes increasing inefficient as uncoordinated concurrent transmissions cause excessive collisions. Therefore, link scheduling mechanisms determine a subset of links to activate while silencing others, which are generally preferred for interference management. The link scheduling problem is typically formulated as a maximum weighted independent set (MWIS) problem on a conflict graph, where vertices represent communication links and edges capture interference relationships [3], [4]. Although MWIS-based scheduling can achieve throughput optimality under ideal conditions [5], the problem is NP-hard and solving it becomes difficult in dense and dynamic ad-hoc networks. To address this, various sub-optimal approximations have been proposed, including distributed greedy schedulers [6] and iterative fractional programming [7]. While these methods offer considerable feasibility, some of them rely on numerous inter-node interactions, others require accurate or global channel state information (CSI) or incur unaffordable computation, imposing heavy energy burdens on resource-limited UAVs.

Recently, learning-based approaches have emerged as a promising alternative [4], [8]–[10]. In particular, graph neural networks (GNNs) align with the interference graph structure of scheduling MWIS problems, and can exploit localized message passing to learn effective representations from coarse observations. Nevertheless, existing GNN-based wireless scheduling methods still face limitations in dense UAV swarm networks. Prevalent models often rely on simplified undirected graphs, failing to distinguish heterogeneous interference strengths in dense deployments. Moreover, most approaches adopt spectral graph convolution networks (GCNs) [4], [9], which are restricted to symmetric graph structures and often entail higher coordination overhead, limiting scalability in large, densely connected networks.

To address these gaps, we propose a distributed graph attention networks (GATs)-assisted link scheduling framework for dense UAV swarm ad-hoc networks. Our approach models a weighted directed conflict graph and explicitly maps edge weights into the attention computation, allowing each link to infer dominant interference patterns from low-dimensional local observations. This edge-enhanced attention mechanism allows the scheduling agent to adaptively prioritize dominant interference while suppressing weakly coupled ones. By operating in the spatial domain rather than the spectral domain, GATs perform local aggregation via message passing without requiring global graph processing, better matching the decentralized and dynamic nature of dense FANETs.

The main contributions of this paper can be summarized as follows:

*i)* We construct a quantized and directed conflict graph

model for link scheduling problem in dense UAV swarm networks. By explicitly incorporating interference intensity into directed edge weights, the model captures more accurate features with minimal measurement overhead.

*ii)* We develop an edge-enhanced GAT-based distributed scheduler that learns from low-dimensional local features and extracts network-wide interference awareness through limited message passing.

*iii)* Simulation results demonstrate that our method significantly improves spectrum utilization, converges rapidly with extremely few training samples, and exhibits robust scalability across different network scales, validating its high data efficiency and low computational energy consumption for green UAV communications.

The remainder of this paper is as follows. Section II presents the system model. Section III introduces the GAT-based scheduling framework. Simulation results are given in Section IV, and Section V concludes the paper.

## II. System Model

In this paper, we consider a dense FANET comprised of $N$ single-antenna UAVs, which are densely distributed in a volumetric 3D space. We consider a single-radio single-channel (SRSC) wireless network, where all nodes communicate with their neighbors over the same channel. The air-to-air (A2A) channel between $\mathrm{UAV}_i$ and $\mathrm{UAV}_j$ can be modeled as [11]:

$$h_{i,j} = a_{i,j} g_{i,j}, \quad i,j \in \{1,2,\ldots,N\}, \tag{1}$$

where $a_{i,j} = \sqrt{\lambda_0} d_{i,j}^{-\xi/2}$ represents the large-scale path loss. $\lambda_0$ denotes the path loss at the reference of 1 m, and $d_{i,j}$ is the distance between $\mathrm{UAV}_i$ and $\mathrm{UAV}_j$. $\xi > 0$ is the path loss exponent, which is typically set to 2.2 to 2.7 for A2A channels [12]. The small-scale factor $g_{i,j}$ is assumed to follow Rayleigh fading, satisfying $g_{i,j} \sim \mathcal{CN}(0,1)$.

This SRSC FANET can be expressed as a directed communication graph $\mathcal{G}_c(\mathcal{U},\mathcal{V})$, where $\mathcal{U} = \{\mathrm{UAV}_1,\ldots,\mathrm{UAV}_N\}$. We consider a minimum input sensitivity at the receiver, so that there exists a communication link $v_{i,j} \in \mathcal{V}$ between $\mathrm{UAV}_i$ and $\mathrm{UAV}_j$ only if the signal-to-noise ratio (SNR) meets the threshold $\beta$ [13]:

$$\mathsf{SNR}_{i,j} = \frac{P|h_{i,j}|^2}{N_0} \geq \beta, \tag{2}$$

where each node transmits signals with a constant power $P$, and $N_0$ is the power of the additive white Gaussian noise.

To characterize competition and interference in wireless networks with simultaneous access to the same time-frequency resources, *conflict graphs* are widely employed to depict the interference relationships between communication links. Existing studies are primarily based on two mainstream interference models [14], [15]: the *protocol model* and the *physical model*. In a conflict graph, vertices represent active communication links in the UAV swarm ad-hoc network, while edges indicate the interference relationships of these links. The protocol model employs a binary representation of interference levels. An undirected edge exists between two vertices if the geometric distance or interference exceeds a certain threshold. Although this representation is highly concise and analytically tractable, it fails to capture the continuous and accumulative extent of interference in dense swarm with massive links.

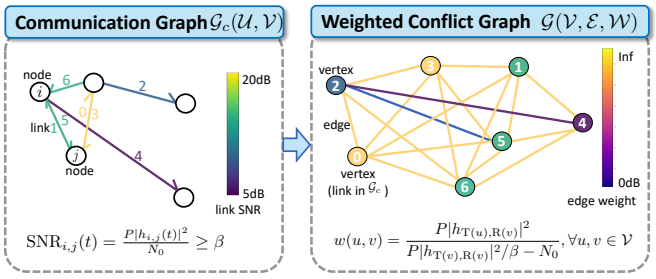



Fig. 1: Mapping from communication graph to directed conflict graph.

To overcome this, we construct a weighted and directed conflict graph $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{W})$ based on the physical interference model to provide a more precise and realistic characterization of the wireless environment. As illustrated in Fig. 1, the vertices of conflict graph $\mathcal{G}(\mathcal{V},\mathcal{E},\mathcal{W})$ are mapped from the link set of the communication graph $\mathcal{G}_c(\mathcal{U},\mathcal{V})$. For any distinct pair of links $(u,v) \in \mathcal{V}$, if $P|h_{\mathrm{T}(u),\mathrm{R}(v)}|^2 \geq \frac{P|h_{\mathrm{T}(v),\mathrm{R}(v)}|^2}{\beta} - N_0$ [15], there is a directed edge $e(u,v) \in \mathcal{E}$. Specifically, the weight of edge $e(u,v)$ is given by $w(u,v) \in \mathcal{W}$ [15], quantifying the specific interference degree:

$$w(u,v) = \frac{P|h_{\mathrm{T}(u),\mathrm{R}(v)}|^2}{P|h_{\mathrm{T}(v),\mathrm{R}(v)}|^2/\beta - N_0}, \forall u,v \in \mathcal{V}, \tag{3}$$

where $\mathrm{T}(v)$ and $\mathrm{R}(v)$ represent the transmitter and receiver of $v$. The quantized weights $w(u,v)$ can be approximated via short-term local measurements such as SNR or received signal strength (RSS) information, eliminating dependence on instantaneous global CSI measurements.

Let $\boldsymbol{\pi} = [\pi_v]_{v\in\mathcal{V}} \in \{0,1\}$ denote the scheduling decision vector for links $v \in \mathcal{V}$. The link $v$ is scheduled for transmission when its corresponding variable $\pi_v$ is set to 1, and silenced otherwise. Treating inter-link interference as noise, the achievable rate of each link $v$is given by:

$$R_v = B\log_2\left(1 + \frac{\pi_v P|h_{\mathrm{T}(v),\mathrm{R}(v)}|^2}{\sum_{u\in\mathcal{N}(v)} \pi_u P|h_{\mathrm{T}(u),\mathrm{R}(v)}|^2 + N_0}\right) \tag{4}$$

where $B$ is the bandwidth, and $\mathcal{N}(v)$ denotes the incoming neighbor set of $v$ in the conflict graph $\mathcal{G}$. Finally, to optimize the spectrum efficiency in dense FANET, we formulate the scheduling problem as a network sum-rate maximization:

$$\mathcal{P}: \underset{\{\pi_v\in\{0,1\}|\forall v\in\mathcal{V}\}}{\arg\max} \sum_{v\in\mathcal{V}} \pi_v \cdot R_v. \tag{5}$$

Note that this problem can be extended to incorporate quality of service (QoS) requirements by introducing flexible utility function such as queue length or priority of service to problem formulation.

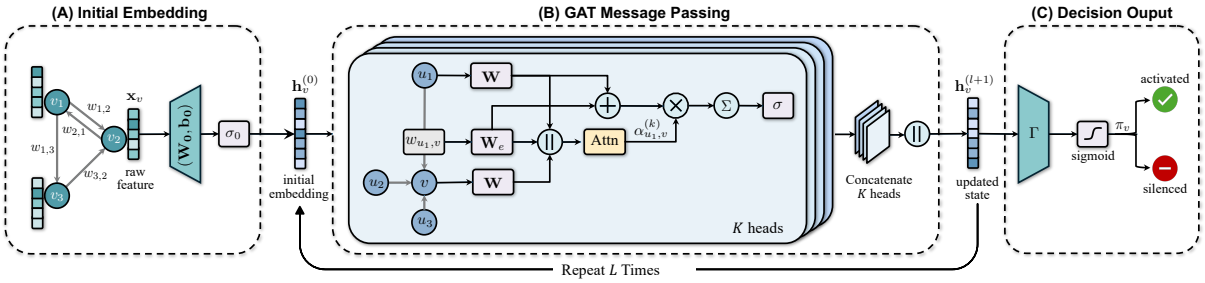



Fig. 2: Illustration of the GAT-assisted distributed scheduling framework.

## III. Distributed Link Scheduling via GATs

### A. Framework Design and Feature Embedding

In a dense UAV swarm, conflict graphs exhibit extremely high vertex degrees, introducing substantial computational challenges for spectral-based GNN scheduling schemes. To address the lightweight requirement in distributed large-scale swarms, we adopt a spatial-based graph learning framework. GAT mechanisms perform message passing by directly aggregating information from neighbors in the graph domain, which enables each UAV link to make decisions solely dependent on localized observations.

We employ a centralized training and distributed execution paradigm to deploy our proposed scheduling scheme. As illustrated in Fig. 2, during the offline training phase, the model exploits global network snapshots to learn generalized topological interference patterns. Once trained, it is deployed on individual UAV nodes in a fully distributed manner. Each node operates online scheduling with low-dimensional local observations, eliminating the need for real-time acquisition of global CSI and prohibitive inter-node overhead.

Specifically, we first leverage a *struct2vec* graph embedding [10] to construct state features. Each vertex $v \in \mathcal{V}$ constructs a raw feature vector $\mathbf{x}_v = [\mathsf{SNR}_v]$, which is then mapped into low-dimensional embeddings via a multi layer perceptron (MLP), serving as initial vertex representations for the subsequent GAT message passing:

$$\mathbf{h}_v^{(0)} = \sigma_0\left(\mathbf{W}_0\mathbf{x}_v + \mathbf{b}_0\right), \tag{6}$$

where $\mathbf{h}_v^{(0)}$ denotes the initial embedding of $v$, $\sigma_0(\cdot)$ is an activation function, and $\{\mathbf{W}_0, \mathbf{b}_0\}$ are learnable parameters.

### B. Edge-Enhanced Attention Mechanism

To achieve low complexity distributed scheduling, we propose an iterative message-passing GAT mechanism. By aggregating information within local neighborhoods over finite iterations, each vertex progressively expands its effective receptive field to implicitly capture global interference patterns.

Let $\mathbf{h}_v^{(l)}$ denote the feature vector of vertex $v$ at the $l$-th GAT layer. At each iteration, the GAT layer assembles information from the neighbor set $\mathcal{N}(v)$ to update vertex representation. To capture asymmetric interference relationships, we incorporate the physical edge weights $w_{u,v}$ into the attention coefficients. The correlation score $e_{u,v}^{(l)}$ is evaluated as:

$$e_{u,v}^{(l)} = \mathrm{LeakyReLU}\left(\mathbf{a}^{\mathrm{T}}\left[\mathbf{W}\mathbf{h}_v^{(l)} \| \mathbf{W}\mathbf{h}_u^{(l)} \| \mathbf{W}_{\mathrm{e}} w_{u,v}\right]\right), \tag{7}$$

where $\mathbf{a}^{\mathrm{T}}$ is the attention weight vector, $\{\mathbf{W}, \mathbf{W}_{\mathrm{e}}\}$ transform vertex and edge features. The scores are normalized via softmax function to obtain the attention coefficient $\alpha_{u,v}^{(l)}$:

$$\alpha_{u,v}^{(l)} = \frac{\exp\left(e_{u,v}^{(l)}\right)}{\sum_{k\in\mathcal{N}(v)} \exp\left(e_{k,v}^{(l)}\right)}. \tag{8}$$

Compared with traditional unweighted models, explicitly incorporating directed edge features ensures the heterogeneous interference relationships precisely reflected in neural embeddings.

To capture diverse interference dynamics such as varying geographic distances and rapid small-scale fading, we adopt multi-head attention. By executing $K$ independent attention heads in parallel, the model projects the vertex features into multiple distinct representation subspaces. Their outputs are concatenated to form the final update for each iteration:

$$\mathbf{h}_v^{(l+1)} = \Big\|_{k=1}^{K} \sigma\left(\sum_{u\in\mathcal{N}(v)} \alpha_{u,v}^{(k)}\left(\mathbf{W}^{(k)}\mathbf{h}_u^{(l)} + \mathbf{W}_{\mathrm{e}}^{(k)} w_{u,v}\right)\right), \tag{9}$$

where the superscript $(k)$ denotes parameters of the $k$-th head.

### C. Unsupervised Learning and Final Scheduling Decision

Dense conflict graphs tend to be fully connected, so only a few message-passing iterations are adequate [10]. In practice, $L = 2$ is sufficient to capture dominant local interference without incurring over-smoothing.

After $L$ rounds, the final scheduling probability $\pi_v \in (0,1)$ is generated by mapping the final embedding $\mathbf{h}_v^{(L)}$ through an output MLP combined with a sigmoid activation:

$$\tilde{\pi}_v = \mathrm{sigmoid}\left(\Gamma(\mathbf{h}_v^{(L)})\right) \in (0,1). \tag{10}$$

To preserve differentiability, scheduling variables are relaxed to soft continuous $\tilde{\pi}_u$ in the training process. The resulting continuous achievable rate is:

$$\tilde{R}_v = B\log_2\left(1 + \frac{\tilde{\pi}_v P|h_{\mathrm{T}(v),\mathrm{R}(v)}|^2}{\sum_{u\in\mathcal{N}(v)} \tilde{\pi}_u P|h_{\mathrm{T}(u),\mathrm{R}(v)}|^2 + N_0}\right). \tag{11}$$

**Algorithm 1** Distributed GAT-assisted Link Scheduling

**Require:** Conflict graph $\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathcal{W})$, local node features $\{\mathbf{x}_v\}_{v\in\mathcal{V}}$, layer count $L$, attention heads $K$.
**Ensure:** Discrete link activation decisions $\pi_v \in \{0, 1\}$.

**Initialization Phase:**
Extract initial embeddings: $\mathbf{h}_v^{(0)} = \sigma_0(\mathbf{W}_0\mathbf{x}_v + \mathbf{b}_0)$.
**Message Passing Phase:**
**for** layer $l = 0$ to $L-1$ **do**
  **for** each vertex $v \in \mathcal{V}$ **in parallel do**
    **for** each head $k = 1$ to $K$ **do**
      Compute attention score $e_{u,v}^{(k)}$ and normalized weight $\alpha_{u,v}^{(k)}$ for neighbors $u \in \mathcal{N}(v)$ via Eq. (8).
      Aggregate: $\mathbf{z}_v^{(k)} = \sum_u \alpha_{u,v}^{(k)} \left(\mathbf{W}^{(k)}\mathbf{h}_u + \mathbf{W}_{\mathrm{e}}^{(k)} w_{u,v}\right)$.
    **end for**
    Update state: $\mathbf{h}_v^{(l+1)} = \Big\|_{k=1}^{K} \sigma(\mathbf{z}_v^{(k)})$.
  **end for**
**end for**
**Decision Output Phase:**
**for** each vertex $v \in \mathcal{V}$ **in parallel do**
  Compute soft probability: $\tilde{\pi}_v = \mathrm{sigmoid}\left(\Gamma(\mathbf{h}_v^{(L)})\right)$.
  Binarize decision: $\pi_v = \mathbb{1}(\tilde{\pi}_v \geq 0.5)$.
**end for**

We adopt an unsupervised learning paradigm where the GNN is trained end-to-end to minimize the negative expected sum rate:

$$\mathcal{L}(\boldsymbol{\theta}) = -\sum_{v\in\mathcal{V}} \tilde{\pi}_v \cdot \tilde{R}_v. \tag{12}$$

During the online deployment phase, the soft output $\tilde{\pi}_v$ is binarized to yield the final discrete scheduling decisions $\pi_v$. The complete inference process executed at each node is summarized in Algorithm 1.

*Remark 1 (On Loss Function Design and Training Stability):* A critical challenge in training unsupervised GNNs is convergence stability. Since the loss function in Eq. (12) applies symmetrically to all links, in our early experiments we observed that gradients frequently push all nodes to act greedily, causing a model collapse into an all-ones (full transmission collision) state.

To actively guide gradients, inspired by existing literature [9], we first introduce a sparsity penalty, which merely forced the network to toggle between all-ones and all-zeros collapse. Subsequently, we innovatively designed a joint sparsity and conflict penalty to guide distributed nodes to learn a collectively optimal scheduling strategy and achieve uniform sparsification:

$$\mathcal{L}_{\mathrm{reg}}(\boldsymbol{\theta}) = -\sum_{v\in\mathcal{V}} \tilde{\pi}_v \cdot \tilde{R}_v + \lambda_1 \frac{1}{|\mathcal{V}|} \sum_{v\in\mathcal{V}} \tilde{\pi}_v + \lambda_2 \sum_{e(u,v)\in\mathcal{E}} w_{u,v} \tilde{\pi}_u \tilde{\pi}_v. \tag{13}$$

Empirically, we observed that this penalty indeed acted as a reliable "safety guard", successfully averting model collapse at borderline learning rates (e.g., around LR $= 3 \times 10^{-4}$).

However, rigorous ablation studies revealed a deeper insight: while these penalties are theoretically appealing for guiding gradients, the fundamental trigger for trivial collapse in our architecture is gradient explosion within the highly non-convex fractional landscape. By adopting a strictly controlled learning rate regime (LR $= 10^{-4}$), the GAT can achieve robust training stability and inherently learn the competitive inhibition via a pure sum-rate objective. This concise design frees the system from hyperparameter tuning associated with auxiliary penalties.

## IV. Simulation Results

In this section, we evaluate the performance of the proposed GAT-based distributed scheduling scheme. We first provide scheduling performance in terms of network throughput, and then investigate the convergence behavior, data efficiency and scalability performance of the proposed learning framework.

### A. Implementation Details

We consider a dense UAV swarm network deployed within a $1000\,\mathrm{m} \times 1000\,\mathrm{m} \times 100\,\mathrm{m}$ airspace. To simulate a realistic dense deployment, we vary the number of UAV nodes $N$ from 20 to 50, ensuring a minimum separation distance of 20 m to avoid physical collisions.

The proposed model is implemented using PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU. We adopt the Adam optimizer with a learning rate of $10^{-4}$ to ensure stability. The detailed simulation and parameter settings are summarized in Table I.[1]

TABLE I: Simulation Parameters

| Parameters | Values |
|---|---|
| *Network Environment* | |
| Number of UAV links ($N$) | 20 / 30 / 40 / 50 |
| Area Size | $1000 \times 1000 \times 100$ m$^3$ |
| UAV Altitude | $[50, 150]$ m |
| Minimum inter-UAV distance | 20 m |
| Transmit Power ($P_0$) | 10 dBm |
| Carrier Frequency ($f_c$) | 2.4 GHz |
| Noise Power Density | $-169$ dBm/Hz |
| Path Loss Exponent ($\xi$) | 2.5 |
| Connectivity SNR Threshold ($\beta$) | 5 dB |
| *Model Architecture* | |
| MLP Hidden Dims for Embedding | [32, 32] |
| Embedding Output Dim | 32 |
| GAT Layers ($L$) | 4 |
| Attention Heads ($K$) | 4 |
| GAT Output Dim of All Heads | 48 |
| Scheduling MLP Hidden Dim | 32 |
| *Training Configuration* | |
| Optimizer | Adam (LR=$10^{-4}$) |
| Epochs | 30 |
| Training / Validation / Test Samples | 300 / 128 / 128 |

### B. Network Throughput Performance

To evaluate the performance of our proposed GAT-based scheduler, we illustrate the network throughput performance defined in Eq. (5) comparing with the following benchmarks:

[1]The simulation source code of this paper is available at https://github.com/qwweezeng/GNN_Scheduling.

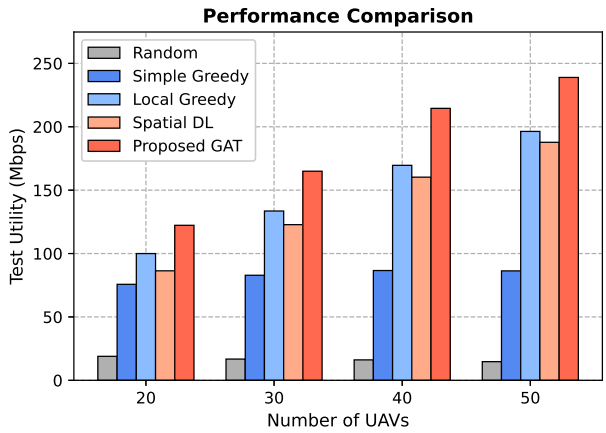



Fig. 3: Utility performance compared with benchmarks.

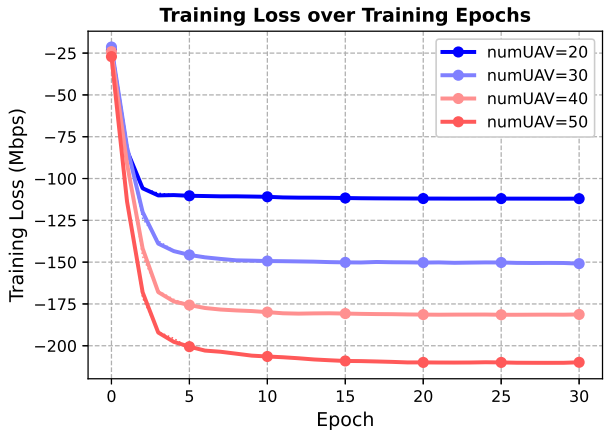



Fig. 4: Training loss versus epochs under different densities.

- **Random Activation (Random):** Each link is independently activated with a fixed probability $p = 0.05$.
- **Signal-Strength Greedy (Simple Greedy):** A fixed proportion (set to $10\%$) of links with the highest local SNR are activated, ignoring potential mutual interference.
- **Interference-Aware Greedy (Local Greedy):** A simplified version of the distributed greedy scheduler [6], which iteratively activates the link with highest SNR, while all its conflicting neighbors are removed from candidates.
- **Spatial Deep Learning (Spatial DL) [9]:** A GCN-based scheduling paradigm over undirected interference graph.

Fig. 3 illustrates the test utility as the network density increases from $N = 20$ to $N = 50$ UAVs. Our proposed GAT consistently outperforms all benchmark schemes. The *Random* and *Simple Greedy* schemes exhibit nearly constant throughput across different network sizes, since their activation policies are static and do not adapt to the complex interference structure. *Local Greedy* achieves higher throughput by explicitly avoiding strong interference through conflict-aware link selection. However, its reliance on iterative neighborhood coordination and sequential signaling overhead limit its practicality.

The *Spatial DL* scheduler benefits from learning-based feature extraction. Nevertheless, its performance gap with the proposed approach becomes more pronounced as the network density increases. This can be attributed to its GCN-style spatial convolution, which relies on aggregating information over a fixed graph structure and implicitly assumes a predefined link set. Moreover, the global convolution required by GCN-based models incurs higher computational and signaling overhead, making them less suitable for dense ad-hoc deployments with strict latency and scalability constraints.

### C. Convergence Analysis

To examine training stability and convergence speed, we plot the training loss over epochs under different network densities in Fig. 4. The proposed model exhibits rapid and stable convergence for all density scenarios. The training loss decreases sharply in early phases and reaches a steady state within merely 10 to 15 epochs.

Such fast convergence can be attributed to the proposed interference-aware modeling and the edge-enhanced attention mechanism. By constructing a weighted conflict graph and explicitly incorporating physical edge weights into GAT attention calculation, the model efficiently captures the dominant interference relationships through local message passing. This enables each node to quickly infer the global interference structure from limited neighborhood interactions, thereby radically accelerating the learning of effective, collision-free scheduling policies.

### D. Training Data Efficiency

We further investigate the training data efficiency by exploring model utility with different numbers of training samples. As shown in Fig. 5, our model achieves over $90\%$ of its peak performance with as few as 40 to 60 training samples. The performance gain saturates after the dataset size exceeds 100 samples.

For comparison, existing GNN-assisted works such as *Spatial DL*, typically require up to 800,000 complex geographic layouts [9] to train their models effectively. In contrast, our method significantly reduces the data requirement by over four orders of magnitude. This phenomenon demonstrates the strong inductive bias of our proposed GAT architecture. Instead of memorizing specific channel realizations or instantaneous locations, the model inherently learns generalizable topological rules for interference management. Consequently, the proposed approach enables extremely lightweight training and rapid deployment, making it highly practical for dynamically varying FANET swarms with limited historical data availability.

### E. Model Scalability Across Network Scales

We further investigate generalization performance of our proposed GAT-based scheduler across varying network scales. To this end, models trained exclusively on datasets with a specific density $N \in \{20, 30, 40, 50\}$ UAV nodes are then

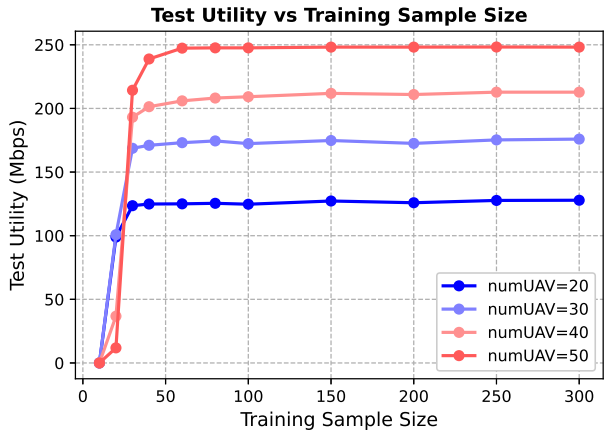



Fig. 5: Utility performance versus the training sample sizes.

TABLE II: Model Scalability Across Network Scales

| **Training Scale** | **Testing Scale** | | | |
|---|---|---|---|---|
| | $N = 20$ | $N = 30$ | $N = 40$ | $N = 50$ |
| $N = 20$ | 24.46 | 32.89 | 42.71 | 47.48 |
| $N = 30$ | 23.70 | 32.99 | 42.89 | 47.74 |
| $N = 40$ | 7.52 | 32.98 | 42.91 | 47.79 |
| $N = 50$ | 5.20 | 32.85 | 42.91 | 47.78 |
| Mixed ($N = 20 \sim 50$) | 24.46 | 32.99 | 42.85 | 47.78 |

cross-evaluated on test sets with mismatched network sizes. As demonstrated in Table II, our proposed scheme demonstrates robust generalization capability.

Notably, models trained on small-scale networks can effectively generalize to large-scale networks. The model trained on $N = 20$ nodes achieves over 99% of the optimal performance (the benchmark where training and testing sets are consistent) when applied to a more dense testing network with $N = 50$. This indicates that our GAT-based model learns interference management patterns which are robust to network scale and dynamic node density variations.

Conversely, model trained on $N = 50$ large-scale networks suffers a 79% degradation when adapting to sparse $N = 20$ scenarios. This asymmetry can be attributed to the excessive reception fields in dense networks. Models trained in large-scale networks rely on dense neighborhood information, potentially overfitting to extensive interference patterns, making them difficult to adapt to sparser topologies.

To summarize, the results demonstrate a pivotal practical advantage of our scheme: training on small-scale networks is sufficient for deployment on larger-scale scenarios, which can reduce the training overhead while maintaining scheduling performance.

## V. Conclusion

In this paper, we proposed a distributed link scheduling framework for dense UAV swarm networks. By constructing a quantized directed conflict graph, we accurately characterized the asymmetric spatial interference. Based on this model, we proposed an edge-enhanced GAT scheduler, where quantized interference weights are explicitly integrated into the attention mechanism. This design enables individual nodes to effectively perceive global network situational awareness through finite local message passing, significantly accelerating the learning process. Simulation results demonstrated that our approach realizes high spectral efficiency, consistently outperforming given benchmarks. Notably, our model exhibits remarkable data efficiency and structural scalability, achieving rapid convergence with fewer than 100 training samples and seamlessly generalizing from small-scale training to large-scale deployments. This data efficient and low overhead design provides a practical, lightweight and eco-friendly solution for the power-constrained dynamic FANETs with strict latency and scalability constraints.

## References


[1] J. Chen, J. Wang, J. Wang, and L. Bai, "Joint Fairness and Efficiency Optimization for CSMA/CA-Based Multi-User MIMO UAV Ad Hoc Networks," *IEEE J. Sel. Top. Signal Process.*, pp. 1–13, 2024.

[2] A. Jindal and K. Psounis, "On the Efficiency of CSMA-CA Scheduling in Wireless Multihop Networks," *IEEEACM Trans. Netw.*, vol. 21, no. 5, pp. 1392–1406, Oct. 2013.

[3] J. Ni, B. Tan, and R. Srikant, "Q-CSMA: Queue-Length-Based CSMA/CA Algorithms for Achieving Maximum Throughput and Low Delay in Wireless Networks," *IEEEACM Trans. Netw.*, vol. 20, no. 3, pp. 825–836, Jun. 2012.

[4] Z. Zhao, G. Verma, A. Swami, and S. Segarra, "Distributed Link Sparsification for Scalable Scheduling Using Graph Neural Networks," *IEEE Trans. Wirel. Commun.*, vol. 25, pp. 3879–3893, 2026.

[5] L. Tassiulas and A. Ephremides, "Stability properties of constrained queueing systems and scheduling policies for maximum throughput in multihop radio networks," *IEEE Trans. Automat. Control*, vol. 37, no. 12, pp. 1936–1948, Dec. 1992.

[6] C. Joo and N. B. Shroff, "Local greedy approximation for scheduling in multihop wireless networks," *IEEE Trans. Mobile Comput.*, vol. 11, no. 3, pp. 414–426, Mar. 2012.

[7] K. Shen and W. Yu, "FPLinQ: A cooperative spectrum sharing strategy for device-to-device communications," in *2017 IEEE Int. Symp. Inf. Theory ISIT*, Jun. 2017, pp. 2323–2327.

[8] W. Jiang, "Graph-based deep learning for communication networks: A survey," *IEEE Commun. Surveys Tuts.*, Mar. 2022, also arXiv:2106.02533.

[9] W. Cui, K. Shen, and W. Yu, "Spatial Deep Learning for Wireless Scheduling," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1248–1261, Jun. 2019.

[10] M. Lee, G. Yu, and G. Y. Li, "Graph embedding-based wireless link scheduling with few training samples," *IEEE Trans. Wirel. Commun.*, vol. 20, no. 4, pp. 2282–2294, Apr. 2021.

[11] L. Bai, J. Xu, J. Wang, R. Han, and J. Choi, "Efficient Hybrid Transmission for Cell-Free Systems via NOMA and Multiuser Diversity," *IEEE Trans. Mob. Comput.*, vol. 24, no. 4, pp. 3359–3371, Apr. 2025.

[12] T. Liu, Z. Zhang, H. Jiang, Y. Qian, K. Liu, J. Dang, and L. Wu, "Measurement-based characterization and modeling for low-altitude UAV air-to-air channels," *IEEE Access*, vol. 7, pp. 98 832–98 840, Jul. 2019.

[13] J. Wang, L. Bai, Z. Fang, R. Han, J. Wang, and J. Choi, "Age of Information Based URLLC Transmission for UAVs on Pylon Turn," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8797–8809, Jun. 2024.

[14] P. Gupta and P. Kumar, "The capacity of wireless networks," *IEEE Trans. Inf. Theory*, vol. 46, no. 2, pp. 388–404, Mar. 2000.

[15] K. Jain, J. Padhye, V. N. Padmanabhan, and L. Qiu, "Impact of interference on multi-hop wireless network performance," in *Proceedings of the 9th Annual International Conference on Mobile Computing and Networking*. New York, NY, USA: Association for Computing Machinery, Sep. 2003, p. 66–80.